

K-audit — A_{sub}↔Hall dictionary, Phase-2 first wave (L1 keystone + L2 region + L3 chirality + L4 mass framework + #414 v0.2 + #416 lepton flip); verdict CONDITIONAL PASS with 3 findings

Keeper

2026-05-29 Friday EDT

K-audit — Dictionary first wave

Verdict

CONDITIONAL PASS. The dictionary's first wave is internally consistent, well-tier-marked, and honestly isolates the bet to ONE keystone identification (canonical basis element = physical particle with that K-type). Three findings for v0.2 cleanup; none CRITICAL.

Per-deliverable tier-gates

L1 — A_{sub}↔Hall keystone (Lyra_Dictionary_A_sub_to_Hall_v0_1_Keystone.md)

- The keystone — RIGOROUS as rep theory. $U_q(B_2)$ has root system B_2 ; $SO(5) \subset K=SO(5) \times SO(2)$ has root system B_2 . The IDENTIFICATION that this is the SAME B_2 (algebra weight $\leftrightarrow$ K-type $SO(5)$ weight) is the dictionary content; correctly tier-marked as THE bet, isolated and falsifiable (it predicts σ_{BF} and charge from weight, both check on Grace's K-types).
- σ_{BF} derivation (weight parity $\rightarrow$ boson/fermion) — DERIVED modulo the keystone bet. Standard $Spin(5)$ rep theory; verified on lepton (4-spinor, half-integer, FERMION), photon (5, integer, BOSON), gauge (10, integer, BOSON), Higgs (1, integer, BOSON). [\[?\]](#)
- Charge handle ($SO(2)$ weight) — DERIVED-as-handle; full GMN staged for L2.
- Region / chirality / winding STAGED with concrete handles, not hand-waves.

L2 — Region + GMN (Lyra_Dictionary_L2_Region_and_GMN_v0_1.md)

- Geometric structure — RIGOROUS. Shilov boundary $S = (S^4 \times S^1)/Z_2$; $\dim 5 = n_C$; K-stabilizer $M = SO(4) = SU(2)_L \times SU(2)_R$; dim check $11-6=5$ [\[?\]](#).
- Region DERIVED (geometric, modulo keystone) — Hardy(Shilov, $SO(4)$) vs Bergman(bulk, $SO(5)$). Two physics facts from one geometric fact: lepton-on-Shilov AND lepton colorlessness, because the Shilov stabilizer $SO(4)$ has no color structure. Clean.
- GMN LOCATED, NOT ASSERTED. $T_3 \in SU(2)_L$, Y from $SU(2)_R + SO(2)$; exact embedding coefficients pinned-not-asserted with built-in acceptance test (must reproduce $\sin^2 \theta_W = 2/9$). Exemplary Source-Verification discipline.
- Chirality handle surfaced (boundary breaking $SU(2)_L \leftrightarrow SU(2)_R$) — fed into L3.

L3 — Chirality (Lyra_Dictionary_L3_Chirality_v0.1.md)

- Already tier-gated in real time. FRAMEWORK-PLUS (geometric mechanism: D_{IV}^5 Hermitian $\rightarrow J = SO(2)$ factor $\rightarrow$ holomorphic Hardy $\rightarrow SU(2)_L$ coupling $\rightarrow$ left-handed weak; right-handed weak singlets). The substrate's *complex orientation* IS parity violation; one $SO(2)$ does four jobs (Hermitian / $U(1)$ charge / Shilov direction / parity) — that unification is structurally real, not rhetorical.
- Per-particle L/R labeling rides on the $SO(5,2) \rightarrow$ Lorentz embedding (Lyra honestly flagged, pinning not asserting). Honest scope.

L4 — Mass framework (in Lyra's L4 v0.1 report)

- FRAMEWORK (sector-dependent mass mechanisms). Lepton at $V_{(1/2,1/2)}$ Casimir $= 5/2 = \rho_1 =$ kernel singularity exponent = “matter sets the unit.” Sector mechanisms: fermion = Casimir/kernel scale; gauge = EWSB (with the derived $\sin^2\theta_W = 2/9 + m_W/m_Z = \sqrt{(g/N_c)}$ ratios); Higgs = VEV; photon = gauge $\rightarrow$ massless; bulk composite = Bergman volume.
- $m_p/m_e = 6\pi^5 = C_2 \cdot \pi^{(n_C)}$: numerically tautological ($C_2=6$, $n_C=5$ — arithmetic identity); the *structural reading* (adjoint Casimir $\times$ Bergman volume = boundary $\rightarrow$ bulk bridge) is FRAMEWORK-PLUS, riding on the mass-mechanism story. Worth foregrounding because the headline could otherwise sound stronger than it is — T187 was already known ($\sim 0.002\%$); what's new is the *organization*, not the numerical match.

#414 v0.2 — Generation knot reframe

- Absorbed Keeper's two-structures burden honestly. Held the gate at “COUNT-NUMBER FORCED / MECHANISM OPEN-WITH-BURDEN” matching the ledger word-for-word; surfaced a candidate *direction* (bulk-gauge for color + cross-cutting spinor-tower for generations) without overstating it as a mechanism. The discipline fired internally on her own “happy unification” wording — exactly right.

#416 v0.1 — Lepton sector per-particle flip

- First concrete row flips ASSIGNED $\rightarrow$ DERIVED. 18 entries; every static axis derived (σ_{BF} , region, chirality, charge, particle/anti) modulo the keystone bet; generation indexed but per #414 open. Photon/Higgs single-particle clean. The dictionary has produced its first per-particle output — milestone-real.

Findings (severity-rated)

Finding 1 — MINOR

L1's “two routes agree on σ_{BF} ” is two CORRELATED routes, not two independent confirmations. The weight-parity route and the R-matrix-braiding route are both manifestations of the *same* $SO(5)$ weight parity (the braiding sign for half-integer weights $= -1$ is a standard QG consequence of weight parity). This is exactly the same Cal #27 / correlated-routes nuance I held on the L1 sector-selection result earlier this session. Lyra's framing “two routes agree” is true; “two *independent* confirmations” would overstate. Severity: MINOR.

Condition 1: L1 (or a v0.2 footnote) clarifies the two routes are correlated, not independent — STRONG agreement (rep-theoretic consistency) but not independent confirmation.

Finding 2 — MINOR

L2's “color requires the full $SO(5)$ ” wording is mis-located in light of Lyra's later finding (#416 batch: $SU(3) \neq SO(5)$ as rank-2 Lie algebras; color CANNOT come from K). The L2 DISTINCTION (Shilov colorless vs bulk colored) stands — leptons on Shilov are colorless because their stabilizer $SO(4)$ doesn't host color, and quarks must be bulk objects. But the *reason* quarks have color isn't “the full $SO(5)$ is available in the bulk” ($SO(5)$ doesn't host $SU(3)$); it's that the bulk non-compact directions of $SO(5,2)$ supply the color mechanism (which is the new explicit Bulk-Color Gate). Severity: MINOR (wording, not derivation).

Condition 2: L2 v0.2 refines the colorlessness derivation: leptons colorless follows from the Shilov stabilizer $SO(4)$ being too small to host any 3-fold color structure; quark color comes from the bulk non-compact mechanism (open gate), not from $SO(5)$ per se.

Finding 3 — MODERATE (*load-bearing, frame-level*)

The “static taxonomy essentially derived” milestone is accurate ONLY IF the keystone bet is foregrounded with every cell-flip. The keystone — *canonical basis element of $U_q(B_2)$ IS a physical particle with that K-type* — is the single bet that carries the entire dictionary; everything that’s “DERIVED” is DERIVED-modulo-this-bet. Lyra states this honestly in L1 §6, but the milestone-language (“static taxonomy essentially derived”) and the lepton-flip headline (“18 entries, every static axis named by the dictionary”) can read in isolation as if the bet has been retired. It hasn’t. Each downstream DERIVED claim is conditional on the keystone, and the keystone is *narrow and testable* (predicts σ_{BF} + charge from K-type weight — and those checked, which is real evidence) but still a bet until the canonical basis itself derives the K-type $\leftrightarrow$ particle correspondence. Severity: MODERATE (frame-level — affects how the program’s state is communicated outward).

Condition 3: Standing rule (Keeper) — every dictionary-derived cell in the Periodic Table carries an explicit “modulo keystone bet” flag, and the headline “static taxonomy derived” travels with “modulo K-types = particles.” Locked into the honest-state ledger; should propagate to Grace’s per-catalog ledger as a one-line note.

Audit hooks status (the standard ones)

1. The keystone identification is rigorous as rep theory [?] — both objects have B_2 ; the IDENTIFICATION is the dictionary content.
2. Each DERIVED axis traces to the keystone + structure, not to hand-assignment [?] — σ_{BF} from weight parity; Charge from $SO(2)$; Region from Shilov stabilizer; Chirality from holomorphy; particle/anti from F-part.
3. STAGED items have concrete handles, not hand-waves [?] — Spin(5) cover (chirality L/R); ρ -vector (region); δ -tower (winding); embedding coefficients (GMN, with acceptance test).
4. The discipline fires internally [?] — Lyra’s #414 v0.2 own-correction; honest tier marking throughout; coefficients pinned-not-asserted; cal #27 firing on “happy unification” wording.

Net

The dictionary’s first wave is dictionary-ratified at this tier. Four of five static-taxonomy axes are DERIVED-modulo-the-one-bet (σ_{BF} , region, chirality, particle/anti); charge is LOCATED with a concrete pin path; the lepton sector per-particle flip is real; the generation gate is held honestly with its burden absorbed. The Phase-2 keystone is producing exactly what the workplan called for. With the three findings addressed (mostly wording + a standing flag), it’s a clean PASS.

The two big remaining open items (bulk-color mechanism + per-particle L/R labeling pin + generation mechanism) are honestly identified and concretely targeted — not buried.

— Keeper, K-audit verdict CONDITIONAL PASS, 2026-05-29.